

Original article

Association between dietary fibre intake and depressive symptoms before and during the COVID-19 pandemic

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SUMMARY

Background: Emerging evidence indicates that dietary fibre may influence mental health via mechanisms related to the gut–brain axis. However, population-level evidence examining the association between dietary fibre intake and depressive symptoms before and during the COVID-19 pandemic remains limited. We examined the associations between energy-adjusted and energy-unadjusted dietary fibre intake and depressive symptoms across these two time periods.

Methods: We analysed data from the National Health and Nutrition Examination Survey (NHANES) conducted before the COVID-19 pandemic (January 2017–March 2020; $n = 7,513$) and during the pandemic (August 2021–August 2023; $n = 4,680$). Depressive symptoms were assessed using the Patient Health Questionnaire-9 (PHQ-9), with scores ≥ 10 indicating depression. Dietary fibre intake (g/day) was assessed using a 24-h dietary recall. Energy-adjusted dietary fibre intake was derived using the residual method to account for total energy intake, thereby isolating the effect of fibre independent of total caloric consumption. Associations between energy-adjusted and energy-unadjusted dietary fibre intake and depressive symptoms were estimated using modified Poisson regression models, adjusting for relevant covariates including sociodemographic characteristics.

Results: The prevalence of depression increased from 8.0% pre-pandemic to 10.1% during the pandemic. Mean dietary fibre intake decreased from 16.52 g/day pre-pandemic to 15.95 g/day during the pandemic. Each additional 5 g of energy-adjusted fibre intake was associated with a 14% lower prevalence of depressive symptoms before the pandemic (adjusted prevalence ratio (aPR) = 0.86; 95% CI: 0.81, 0.92)), whereas the association was attenuated during the pandemic (aPR = 0.98; 95% CI: 0.91, 1.05). In energy-unadjusted analyses, a 1 g/day increase in fibre intake was associated with a 2% lower depression prevalence (aPR = 0.98; 95% CI: 0.97, 0.99) pre-pandemic but was insignificant (aPR = 0.99; 95% CI: 0.98, 1.00) during the pandemic. Gender-stratified analyses revealed that prior to the pandemic, higher energy-adjusted dietary fibre intake was associated with a lower prevalence of depressive symptoms among men (aPR = 0.79; 95% CI: 0.71, 0.88) and women (aPR = 0.91; 95% CI: 0.84, 0.99). During the pandemic, the association was attenuated among men (aPR = 1.04; 95% CI: 0.92, 1.17), whereas a modest protective association persisted among women (aPR = 0.90; 95% CI: 0.81, 0.99).

Abbreviations: ACASI, Audio Computer-Assisted Self-Interview; aPR, Adjusted Prevalence Ratio; BMI, Body Mass Index; HPA, Hypothalamic–pituitary–adrenal; hs-CRP, High-sensitivity C-Reactive Protein; LOWESS, Locally Weighted Scatterplot Smoothing; MCAR, Missing Completely At Random; MEC, Mobile Examination Center; NCHS, National Center for Health Statistics; NHANES, National Health and Nutrition Examination Survey; PHQ-9, Patient Health Questionnaire-9; PIR, Poverty Income Ratio; SCFAs, Short-Chain Fatty Acids.

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Conclusions: Higher energy-adjusted dietary fibre intake was associated with lower depressive symptoms before the pandemic in both men and women. During the pandemic, however, the protective association persisted only among women. These findings suggest that dietary fibre may offer continued mental health benefits for women even under societal stressors, highlighting its potential as a gender-specific nutritional target for depression prevention.

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1. Introduction

Depression is a leading cause of disability worldwide, affecting approximately 332 million people [1]. Its prevalence has increased markedly in recent years, with the COVID-19 pandemic alone contributing an estimated 53 million additional cases of major depressive disorder globally [2]. Given the significant social and economic consequences of depression, identifying modifiable risk and protective factors remains a major public health priority [3,4]. Overall dietary patterns have been consistently linked to the risk of depression and related mental health outcomes. Healthier dietary patterns, such as the Mediterranean diet, which are rich in fruits, vegetables, legumes, and whole grains, are associated with a lower risk of depression [5,6]. On the other hand, dietary patterns high in processed foods, saturated fats, and added sugars are constantly linked to an increased risk of depression and poorer mental health [7,8]. While studying overall dietary patterns is informative, examining individual nutritional components is necessary to identify specific, modifiable targets for intervention. In this context, dietary fibre has emerged as a key dietary factor with potential relevance for the prevention and management of depression [9–11].

Dietary fibre comprises a heterogeneous group of non-digestible carbohydrates and lignin found in plant-based foods, including vegetables, fruits, nuts, and whole grains [12]. It serves as the primary metabolic substrate for commensal gut microbiota, and its fermentation in the colon produces short-chain fatty acids (SCFAs), including butyrate, propionate, and acetate. These metabolites act as key mediators within the microbiota-gut-brain axis, influencing neuroendocrine pathways and systemic inflammation [11,13,14]. In addition to its established cardiometabolic benefits, fibre may influence mental health through modulation of the hypothalamic–pituitary–adrenal (HPA) axis [14–16] and its anti-inflammatory properties [17]. By reducing pro-inflammatory cytokines and enhancing the integrity of the intestinal barrier, adequate fibre intake may prevent the neuroinflammation associated with depression [18]. Furthermore, SCFAs stimulate the production of brain-derived neurotrophic factor (BDNF) and support serotonergic signalling, providing a robust biological link between fibre consumption and mood regulation [19].

Most existing studies, however, were conducted prior to major societal disruptions such as the COVID-19 pandemic. The pandemic introduced widespread stressors, including social isolation, economic instability, and direct viral effects that substantially increased the risk of depression across populations [2]. This period was also characterised by substantial shifts in dietary behaviours, including lower dietary quality, reduced fruit and vegetable consumption, and decreased physical activity, all of which may have influenced fibre intake and its relationship with depression [20]. Emerging evidence also suggests that COVID-19 infection may induce gut dysbiosis and neuroinflammation, potentially disrupting gut–brain communication and attenuating the protective effect of dietary fibre [17,21]. Moreover, increased

stress from economic instability and social isolation could override the modest protective associations of nutritional factors such as dietary fibre on depression [22].

Although evidence from pre-pandemic cohorts demonstrates that higher dietary fibre intake is associated with lower risk of depressive symptoms, no population-level study has directly examined whether this protective association differed during the COVID-19 pandemic compared with the pre-pandemic period. We hypothesised *a priori* that the inverse association between fibre intake and depressive symptoms would be attenuated during the pandemic. This hypothesis is informed by the stress-buffering framework [23,24], which posits that protective lifestyle factors exert their greatest benefit under conditions of moderate stress but may have diminished effects under sustained and severe stress. During the COVID-19 pandemic, widespread stressors, including social isolation, bereavement, economic uncertainty, and fear of infection, may have exceeded the buffering capacity of dietary factors and disrupted protective biological pathways [25,26].

In this study, using the nationally representative National Health and Nutrition Examination Survey (NHANES) data spanning 2017 to 2023, we aimed to (1) examine the association between energy-adjusted dietary fibre intake and depressive symptoms, and (2) assess whether this association differed between the pandemic period and gender.

2. Methods

2.1. Study design and population

We conducted a cross-sectional comparative analysis using two cycles of the NHANES to evaluate whether the association between energy-adjusted dietary fibre intake and depressive symptoms differed before (January 2017–March 2020) and during (August 2021–August 2023) the COVID-19 pandemic. NHANES is a continuous, nationally representative survey of the civilian, non-institutionalised US population conducted by the National Centre for Health Statistics (NCHS). The survey employs a complex, multistage probability sampling design and includes standardised interviews, physical examinations, and laboratory assessments.

For the present analysis, we included individuals aged 20 or older. Participants aged <20 years were excluded to ensure consistency in the assessment of depressive symptoms using an instrument validated for adults. We further excluded individuals with incomplete data on 24-h dietary recall or depressive symptoms (Fig. 1). Moreover, to improve data validity, we excluded participants with implausible total energy intake (<500 kcal or > 5000 kcal per day) or implausible dietary fibre intake (<0 g/day or >100 g/day) [27]. The 2021–2023 NHANES cycle incorporated pandemic-related modifications, including a delayed start (August 2021) and notably lower response rates (interview: 34.6%; examination: 25.7%) [28].

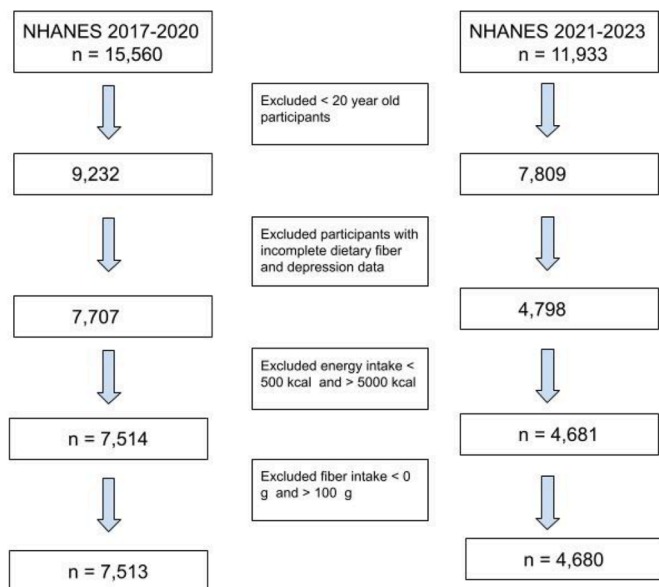


Fig. 1. Flow diagram of participant selection from the NHANES pre-pandemic (January 2017–March 2020) and pandemic (August 2021–August 2023) cycles based on the original unimputed dataset.

To account for the complex survey design and differential non-response across cycles, we used the two-year interview weights for the pre-pandemic cycle (January 2017–March 2020) and the combined interview weights for the pandemic cycle (August 2021–August 2023). For analyses spanning both cycles, we created a combined weight variable by dividing the respective cycle weights by 2, as recommended by the NCHS for analyses pooling two survey cycles. While survey weights adjust for known sources of non-response, the lower response rates during the pandemic cycle may introduce residual selection bias if non-respondents differ from respondents on unmeasured factors relevant to our analysis.

2.2. Exposure variable

The primary exposure in this study was energy-adjusted dietary fibre intake. Total dietary fibre intake (g/day) was obtained from Day 1 24-h dietary recall data collected using the Automated Multiple-Pass Method, a validated five-step interview technique designed to enhance completeness and accuracy of food reporting [29,30]. NHANES dietary interviews capture detailed information on all foods and beverages consumed during the preceding 24-h period (midnight to midnight).

Energy-adjusted dietary fibre intake was derived using the residual method to control for total energy intake, as described by Willett et al. [31]. This approach is essential because total energy intake can confound the relationship between specific nutrients and health outcomes, and because individuals with higher energy consumption naturally consume more fibre. In this procedure, total dietary fibre intake was regressed on total energy intake, and the resulting residuals were used as the energy-adjusted fibre variable. The residual method isolates the component of fibre intake that is independent of total energy intake. A 5 g/day increase in energy-adjusted fibre corresponds to the amount of fibre found in one medium pear (~5.5 g), one medium apple with skin (~4.5 g), two medium Bananas (~6 g), or one cup of broccoli or cooked quinoa (~5 g each) [32]. Practically, this represents replacing energy-dense, low-fibre foods with fibre-rich foods of equivalent caloric value. This transformation was performed prior

to applying survey weights. In this study, “energy-adjusted fibre intake” refers to this derived variable used in all regression analyses, whereas “dietary fibre intake” refers to the unadjusted (absolute) intake values obtained from NHANES [33]. Residualisation was performed prior to applying survey weights because the residual method is a mathematical transformation applied at the individual level to derive a variable that is uncorrelated with total energy intake. Applying weights after this transformation ensures that the derived energy-adjusted fibre variable retains its interpretability as the component of energy-independent fibre intake.

2.3. Outcome variable

The primary outcome of interest was depression symptoms, defined as a Patient Health Questionnaire (PHQ-9) score of ≥ 10 , indicating moderate or severe depressive symptoms. The PHQ-9 is a validated nine-item self-report instrument widely used to screen for and assess the severity of depression [34]. Each item assesses the frequency of depressive symptoms over the past two weeks on a 4-point scale (0 = “not at all” to 3 = “nearly every day”), yielding a total score ranging from 0 to 27. In NHANES, the PHQ-9 is administered at the Mobile Examination Centre (MEC) using Audio Computer-Assisted Self-Interview (ACASI), a method that enhances privacy and improves reporting accuracy [35].

2.4. Covariates

Covariates were selected *a priori* based on established associations with both dietary patterns and depression risk in the literature [36,37]. Covariates included demographic factors (age; gender; race/ethnicity; education; marital status; income category based on poverty income ratio [PIR]: low [<1.3], middle [1.3 – 3.5], high [>3.5]); lifestyle factors (vigorous physical activity defined as ≥ 3 sessions/week of ≥ 20 min or daily ≥ 20 min per WHO guidelines; sleep duration: adequate [≥ 7 h], short [<7 h]; smoking: current yes/no), body mass index (BMI) category (not overweight [<25 kg/m²], overweight/obese [≥ 25 kg/m²]), and prescription medication use (yes/no). High-sensitivity C-reactive protein (hs-CRP) was included as a covariate to account for systemic inflammation, which may partially mediate the effect of dietary fibre on depressive symptoms, as shown in prior NHANES based analyses [38]. hs-CRP levels were classified into low (<1 mg/L), middle (≥ 1 and < 3 mg/L) and high (≥ 3 mg/L) [39]. Alcohol use was defined as consuming at least 12 alcoholic beverages in a year [40].

2.5. Statistical analysis

All analyses were weighted to ensure national representativeness and adjust for the complex sampling design. Sampling weights, clustering, and stratification variables were incorporated according to the NCHS guidelines. Associations between energy-adjusted dietary fibre intake and depressive symptoms were estimated using modified Poisson regression models, producing adjusted prevalence ratios (aPRs) with 95% confidence intervals (CIs). Energy-adjusted dietary fibre intake was modelled per 5 g/day increment to enhance interpretability and maintain consistency with previous literature [41,42]. All analyses were adjusted for a range of covariates described above. All analyses were performed using Stata version 15 (StataCorp, College Station, TX, USA) [43].

2.5.1. Interaction analysis

To examine whether the association between energy-adjusted dietary fibre intake and depression symptoms differed by

pandemic period and gender, we fitted a three-way model. A modified Poisson regression with a log link and robust variance was used, accounting for the complex NHANES survey design, and using the imputed dataset. The model included main effects for energy-adjusted fibre intake (continuous), pandemic period (pre-pandemic versus during pandemic), and gender, as well as all corresponding two-way interactions and the three-way interaction term (fibre $\times$ period $\times$ gender), adjusted for all covariates described above. The interaction model was specified as:

$$\begin{aligned} \log(\text{PR}) = & \beta_0 + \beta_1 (\text{fibre}) + \beta_2 (\text{period}) + \beta_3 (\text{gender}) \\ & + \beta_4 (\text{fibre} \times \text{period}) + \beta_5 (\text{fibre} \times \text{gender}) \\ & + \beta_6 (\text{period} \times \text{gender}) + \beta_7 (\text{fibre} \times \text{period} \times \text{gender}) \\ & + \text{covariates} \end{aligned}$$

where:

- β_0 is the intercept,
- β_1 – β_3 represent the main effects of energy-adjusted fibre intake, period, and gender,
- β_4 – β_6 represent two-way interaction effects, and
- β_7 captures the three-way interaction between energy-adjusted fibre intake, period, and gender

Marginal effects and predicted probabilities were subsequently estimated to aid the interpretation of interaction effects and were graphically presented. The model was estimated using *mi estimate*, *svy: glm*, specifying *family(poisson)* and *link(log)*, and results were reported as aPRs with 95% CIs. All interaction terms were retained regardless of statistical significance, consistent with *a priori* specification.

2.5.2. Subgroup analysis by gender

To evaluate potential effect modification by gender, we conducted subgroup analyses stratified by gender (male vs female) within each pandemic period. This approach allowed us to examine whether the fibre–depression association differed between men and women and whether these associations changed from the pre-pandemic to the pandemic period. Separate modified Poisson regression models were fitted for men and women in the pre-pandemic and pandemic periods. All models incorporated NHANES survey weights and were adjusted for the same set of covariates as in the primary analysis. Results were presented as aPRs with 95% CI.

2.5.3. Missing data and imputation

Patterns of missing data were assessed using Little's Missing Completely at Random (MCAR) test, which indicated data were not missing completely at random. To minimise potential bias, we used multiple imputation by chained equations (MICE) to impute all variables with missing data. These variables included income category (9.9% missing), current smoking status (58.3% missing), alcohol use (13.4% missing), vigorous exercise (12.6% missing), education level (0.07% missing), marital status (0.07% missing), and prescribed medication use (0.11% missing).

2.5.3.1. Imputation model specification. The imputation model was specified as follows. Variables imputed included income category, prescribed medication use, current alcohol use, current smoking status, education level, marital status, and vigorous exercise. Imputation methods were logistic regression for binary variables (current smoking status, current alcohol use, vigorous exercise, prescribed medication use) and multinomial logistic regression for categorical variables with more than two levels (income category,

education level, marital status). The imputation model included all variables from the final analysis model as predictors. To improve the plausibility of the missing at random (MAR) assumption, we also included auxiliary variables: the pandemic period indicator (post_pandemic) and the survey design variables (sdmvpssu, sdmvstra), which were incorporated via the *mi svyset* command. The inclusion of pandemic period is essential as missingness patterns may differ between the pre-pandemic and pandemic periods.

2.5.3.2. Assessment of imputed distributions. To evaluate the credibility of the imputation model, we compared the distribution of smoking status in the observed data with that in the imputed datasets, stratified by period and sex. As shown in [Supplementary Table S1](#), the imputed distributions were highly consistent with the observed data across all strata. For example, the proportion of current smokers among men in the pre-pandemic period was 37.1% (95% CI: 33.3, 41.1) in the observed data and 37.2% (95% CI: 33.7, 40.8) averaged across the 20 imputed datasets. These results provide strong evidence that the imputation model generated plausible values and preserved the underlying relationships between smoking, period, and sex.

2.5.3.3. Number of imputations and diagnostics. We generated 20 imputed datasets, consistent with current methodological recommendations to ensure stable parameter estimates and appropriate standard errors when missingness is moderate or substantial [44,45]. Convergence of the chained equations was evaluated using Monte Carlo error diagnostics. For each parameter, the Monte Carlo error was substantially smaller than its corresponding standard error, indicating stable imputations and adequate convergence of the multiple imputation process. NHANES survey weights, clustering, and stratification were incorporated throughout both the imputation and analysis procedures to preserve the complex survey design.

2.5.4. Sensitivity analyses

We conducted several sensitivity analyses to assess the robustness of our findings. First, we compared results from the multiply imputed analyses with those obtained from complete-case analyses, restricting the sample to participants with complete data on the exposure, outcome, and all covariates. Second, we examined the association using energy-unadjusted dietary fibre intake categorised into quartiles. Quartiles were defined separately within each analytic cycle (pre-pandemic and pandemic) using survey-weighted distributions. Survey-weighted modified Poisson regression models were then fitted for each cycle to estimate aPRs across fibre intake quartiles, using the highest quartile as the reference group. Third, we conducted a sensitivity analysis by excluding smoking status, a variable with a high percentage of missingness.

3. Results

3.1. Participant characteristics

A total of 7,513 participants from the pre-pandemic period (January 2017–March 2020) and 4,680 participants from the pandemic period (August 2021–August 2023) were included in the analysis. Mean dietary fibre intake was slightly lower during the pandemic (15.95 g/day) compared with the pre-pandemic period (16.52 g/day). The demographic profile of participants was broadly comparable across periods, with similar age and racial/ethnic distributions. Educational attainment increased modestly during the pandemic period, whereas marital status and income

distribution largely remained stable. The prevalence of overweight and obesity was high (approximately 74%) in both periods.

During the pandemic, a higher proportion of participants reported adequate sleep, prescription medication use, and engagement in vigorous physical activity, while the prevalence of current smoking was lower. The prevalence of depressive symptoms (PHQ-9 ≥ 10) increased from 8.0% in the pre-pandemic period to 10.1% during the pandemic (Table 1).

As illustrated in Fig. 2, PHQ-9 scores were consistently higher during the pandemic (red line) than in the pre-pandemic period (blue line) across the distribution of dietary fibre intake.

3.2. Association between energy-adjusted and energy-unadjusted dietary fibre intake and depressive symptoms before and during the COVID-19 pandemic

Each 5 g/day increase in energy-adjusted dietary fibre intake was associated with a lower prevalence of depressive symptoms in the pre-pandemic period (aPR = 0.86; 95% CI: 0.81, 0.92). During the pandemic, this association was attenuated, with the estimate shifting towards the null (aPR = 0.98; 95% CI: 0.91, 1.05) (Table 2).

A complementary analysis using energy-unadjusted (absolute) dietary fibre intake (g/day) is presented in Table 3. In the pre-pandemic period, each 1 g/day increase in absolute fibre intake

was associated with a 2% lower prevalence of depressive symptoms (aPR = 0.98; 95% CI: 0.97, 0.99). During the pandemic, the association was attenuated, with estimates centred around the null (aPR = 0.99; 95% CI: 0.98, 1.00). Scaling these estimates to a 5 g/day increment for comparison with the primary analysis yields an aPR of approximately 0.90 (0.98⁵), which is consistent with the primary energy-adjusted estimate of 0.86 (95% CI: 0.81, 0.92).

3.3. Interaction analysis

Table 4 presents the results of the three-way interaction model. Higher energy-adjusted dietary fibre intake was independently associated with a lower prevalence of depressive symptoms (aPR = 0.80; 95% CI: 0.72, 0.89). However, both being in the pandemic period and being female were associated with a higher prevalence of depressive symptoms (aPR = 1.57; 95% CI: 1.23, 2.00 and aPR = 1.46; 95% CI: 1.07, 1.99, respectively).

A stronger interaction between energy-adjusted dietary fibre intake and pandemic period was observed (aPR = 1.33; 95% CI: 1.13, 1.55). The interaction between energy-adjusted dietary fibre and gender was modest (aPR = 1.13; 95% CI: 0.99, 1.29). The three-way interaction (fibre $\times$ pandemic period $\times$ gender) suggested variation in the association across subgroups (aPR = 0.73; 95% CI: 0.58, 0.91) (Table 4).

Table 1

Weighted sociodemographic, behavioural, and clinical characteristics of US adults before (January 2017–March 2020) and during (August 2021–August 2023) the COVID-19 pandemic (weighted %), NHANES.

Variable	Category	Pre-pandemic (n = 7,513)	Pandemic (n = 4,680)
Dietary fibre ^a (g)	Mean (95% CI)	16.52 [15.92, 17.13]	15.95 [15.25, 16.65]
Gender	Male	47.7	48.2
	Female	52.3	51.8
Age category	20–34	27.4	26.2
	35–49	24.4	24.6
	50–64	26.2	26.4
	65+	22.0	22.7
Race	Mexican American	8.3	7.2
	Other Hispanic	7.7	9.6
	Non-Hispanic White	62.7	61.6
	Non-Hispanic Black	11.3	11.3
	Other Race/Multi-racial	10.0	10.4
Education level	Less than 9th grade	3.2	3.0
	9–11th grade	7.0	5.6
	High school graduate	27.4	24.2
	Some college/AA degree	30.6	29.5
	College graduate+	31.8	37.7
Marital status	Married/Partner	61.8	60.1
	Widowed/Divorced	18.7	19.2
	Never married	19.5	20.7
Income category	Low income	22.7	21.3
	Middle income	12.1	13.9
	High income	65.3	64.8
BMI	Not overweight/obese	25.4	25.9
	Overweight/obese	74.6	74.1
Sleep duration	Short (<7 h)	23.4	20.5
	Adequate (≥ 7 h)	76.6	79.5
Prescribed medications	No	40.3	34.3
	Yes	59.7	65.7
Vigorous exercise*	No	91.5	79.6
	Yes	8.5	20.4
Alcohol use	No	43.1	41.9
	Yes	56.9	58.2
Current smoker	No	62.6	64.3
	Yes	37.4	35.7
hs-CRP category	Low	29.3	30.4
	Average	32.0	31.4
	High	38.7	38.2
Depression	No	92.0	89.9
	Yes (PHQ-9 score ≥ 10)	8.0	10.1

^a Energy-adjusted; * WHO definition: ≥ 3 sessions/week of ≥ 20 min or daily ≥ 20 min; hs-CRP: High-sensitivity C-Reactive Protein; AA: Associate of Arts; NHANES: National Health and Nutrition Examination Survey.

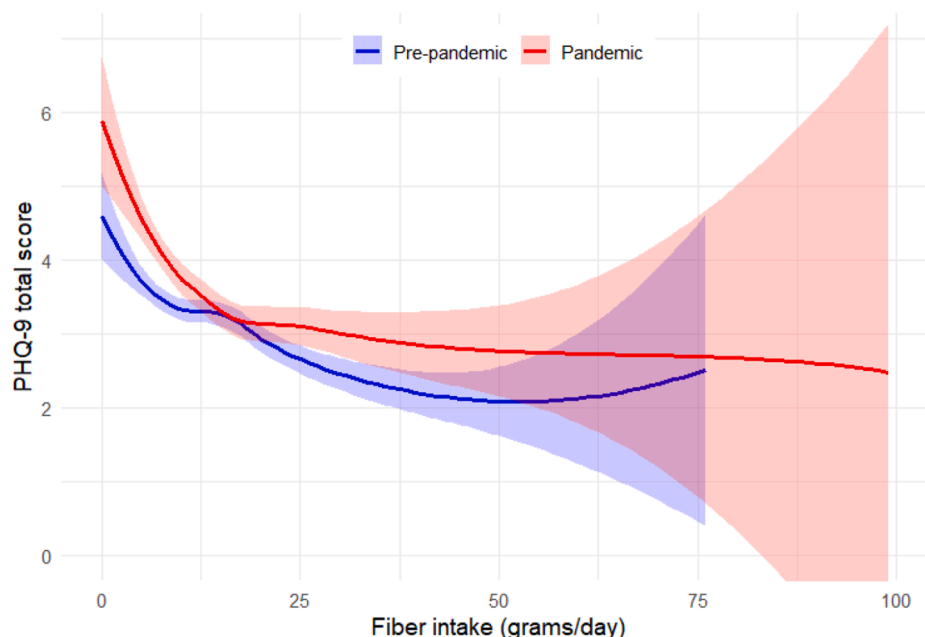


Fig. 2. Locally weighted scatterplot smoothing (LOWESS) of Patient Health Questionnaire (PHQ-9) total score by dietary fibre intake (g/day) before and during the COVID-19 pandemic, NHANES 2017–2023 (with 95% confidence bands).

Table 2

Adjusted prevalence ratios (aPR) for depressive symptoms according to energy-adjusted dietary fibre intake (per 5 g/day) and covariates before and during the COVID-19 pandemic, NHANES.

Variable	Category	Pre-pandemic (n = 7,513)	p	Pandemic (n = 4,680)	p
		aPR (95% CI)		aPR (95% CI)	
Energy-adjusted dietary fibre intake	per 5 g increase	0.86 (0.81, 0.92)	<0.001	0.98 (0.91, 1.05)	0.484
Gender (ref = male)	Female	1.27 (0.96, 1.67)	0.087	1.25 (0.98, 1.61)	0.073
Age category (ref = 20–34)	35–49	1.05 (0.79, 1.41)	0.713	0.72 (0.51, 1.03)	0.066
	50–64	1.02 (0.77, 1.35)	0.907	0.62 (0.46, 0.83)	0.004
	65+	0.62 (0.42, 0.91)	0.018	0.43 (0.30, 0.62)	<0.001
					0.64
Race (ref = Non-Hispanic White)	Mexican American	0.96 (0.58, 1.59)	0.861	0.91 (0.61, 1.37)	0.64
	Other Hispanic	1.19 (0.89, 1.59)	0.233	0.85 (0.61, 1.18)	0.303
	Non-Hispanic Black	0.79 (0.59, 1.05)	0.104	0.68 (0.44, 1.05)	0.079
	Other Race/Multi-Racial	0.99 (0.71, 1.38)	0.958	0.75 (0.52, 1.08)	0.112
Education level (ref = College graduate or above)	Less than 9th grade	1.48 (0.86, 2.53)	0.146	1.33 (0.66, 2.69)	0.392
	9–11th grade (no diploma)	1.54 (0.98, 2.44)	0.062	1.18 (0.58, 2.38)	0.624
	High school graduate	1.52 (0.96, 2.41)	0.074	1.14 (0.83, 1.56)	0.394
	Some college/AA degree	1.55 (0.93, 2.59)	0.086	1.41 (0.99, 2.00)	0.055
Marital status (ref = Married/living with partner)	Widowed/Divorced/Separated	1.38 (1.09, 1.75)	0.01	1.44 (1.00, 2.09)	0.051
	Never married	1.39 (1.01, 1.92)	0.046	1.58 (1.13, 2.20)	0.011
Income category (ref = High income)	Low income	1.91 (1.49, 2.46)	<0.001	1.59 (1.26, 1.99)	0.001
	Middle income	1.84 (1.29, 2.62)	0.002	1.17 (0.85, 1.61)	0.304
BMI overweight/obese (ref = No)	Overweight/obese	0.91 (0.68, 1.22)	0.523	1.27 (0.89, 1.80)	0.174
Sleep duration (ref = Not adequate)	Adequate sleep (≥ 7 h)	0.83 (0.67, 1.03)	0.091	0.61 (0.50, 0.75)	<0.001
Using prescribed medications (ref = No)	Yes	2.77 (1.94, 3.94)	<0.001	1.75 (1.39, 2.22)	<0.001
Vigorous exercise (ref = No)	Yes	0.59 (0.25, 1.38)	0.21	0.76 (0.45, 1.29)	0.267
Alcohol use (ref = No)	Yes	0.72 (0.58, 0.88)	0.003	0.83 (0.64, 1.08)	0.159
Current smoker (ref = No)	Yes	1.43 (1.04, 1.96)	0.028	1.29 (0.88, 1.90)	0.152
hs-CRP category (ref = Low)	Average	1.12 (0.90, 1.41)	0.298	0.64 (0.42, 0.99)	0.044
	High	1.24 (0.90, 1.70)	0.18	0.94 (0.66, 1.33)	0.703

*hs-CRP: high-sensitivity C-Reactive Protein, ref = reference category; AA: Associate of Arts; CI: Confidence Interval; NHANES: National Health and Nutrition Examination Survey; A 5 g/day increase in energy-adjusted dietary fibre is approximately equivalent to the fibre content of one medium pear, one medium apple with skin, two bananas, or one cup of chopped broccoli or cooked quinoa [32].

Figure 3 further illustrates these differences using marginal effects plots. Women had higher predicted depression rates than men at nearly all fibre levels, both before and during the pandemic. The predicted probability of depressive symptoms increased during the pandemic for both men and women. The inverse association between energy-adjusted fibre intake and depressive

symptoms was evident among men in the pre-pandemic period. While the marginal plot suggests a slight negative slope for men during the pandemic, the corresponding statistical estimate was weak (aPR = 1.04; 95% CI: 0.92–1.17; Table 5), highlighting that marginal plots are intended for visual interpretation and statistical estimates should be interpreted as the primary evidence.

Table 3

Adjusted prevalence ratios for depressive symptoms according to energy-unadjusted (absolute) dietary fibre intake (per 1 g/day) before and during the COVID-19 pandemic, NHANES.

Variable	Category	Pre-pandemic		Pandemic	
		aPR (95%CI)	p-value	aPR (95%CI)	p-value
Energy-unadjusted dietary fibre intake	Per 1 g/day increment	0.98 (0.97, 0.99)	0.004	0.99 (0.98, 1.00)	0.152
Gender (ref = male)	Female	1.17 (0.89, 1.54)	0.237	1.22 (0.94, 1.59)	0.12
Race (ref = Non-Hispanic White)	Mexican American	0.90 (0.54, 1.51)	0.684	0.92 (0.62, 1.37)	0.66
	Other Hispanic	1.13 (0.84, 1.53)	0.402	0.85 (0.61, 1.17)	0.29
	Non-Hispanic Black	0.79 (0.59, 1.04)	0.094	0.68 (0.44, 1.05)	0.076
	Other/Multi-Racial	0.96 (0.69, 1.34)	0.795	0.76 (0.53, 1.08)	0.113
	Less than 9th grade	1.47 (0.85, 2.57)	0.161	1.32 (0.66, 2.64)	0.397
Education level (ref = College graduate or above)	9–11th grade (no diploma)	1.64 (1.03, 2.61)	0.038	1.16 (0.57, 2.38)	0.656
	High school graduate	1.59 (0.98, 2.58)	0.059	1.12 (0.82, 1.54)	0.444
	Some college/AA degree	1.61 (0.95, 2.73)	0.073	1.39 (0.98, 1.99)	0.065
	35–49	1.05 (0.78, 1.41)	0.747	0.73 (0.52, 1.03)	0.07
Age category (ref = 20–34)	50–64	0.99 (0.74, 1.32)	0.929	0.62 (0.46, 0.82)	0.003
	65+	0.58 (0.39, 0.86)	0.009	0.43 (0.30, 0.62)	<0.001
	Widowed/Divorced/Separated	1.39 (1.10, 1.77)	0.009	1.44 (1.00, 2.08)	0.05
Marital status (ref = Married/living with partner)	Never married	1.39 (1.00, 1.92)	0.048	1.57 (1.13, 2.19)	0.011
	Low income	1.93 (1.49, 2.48)	<0.001	1.58 (1.26, 1.98)	0.001
Income category (ref = High income)	Middle income	1.84 (1.29, 2.63)	0.002	1.17 (0.85, 1.61)	0.311
	Overweight/obese	0.92 (0.68, 1.23)	0.557	1.26 (0.88, 1.79)	0.186
BMI overweight/obese (ref = No)	Adequate sleep	0.82 (0.66, 1.03)	0.084	0.61 (0.50, 0.75)	<0.001
Sleep duration (ref = Not adequate)	Yes	2.78 (1.95, 3.97)	<0.001	1.75 (1.39, 2.22)	<0.001
Using prescribed medications (ref = No)	Yes	0.58 (0.25, 1.36)	0.202	0.76 (0.45, 1.29)	0.268
Vigorous exercise (ref = No)	Yes	0.72 (0.59, 0.88)	0.003	0.83 (0.64, 1.08)	0.151
Alcohol use (ref = No)	Yes	1.45 (1.06, 1.99)	0.023	1.29 (0.88, 1.90)	0.155
Current smoker (ref = No)	Average	1.12 (0.89, 1.40)	0.317	0.64 (0.42, 0.98)	0.042
hs-CRP category (ref = Low)	High	1.24 (0.90, 1.71)	0.176	0.94 (0.66, 1.33)	0.687

*aPR: adjusted Prevalence Ratio; hs-CRP: high-sensitivity C-Reactive Protein; ref = reference category; AA: Associate of Arts; CI: Confidence Interval; NHANES: National Health and Nutrition Examination Survey.

Table 4

Three-way interaction model: adjusted prevalence ratios for depressive symptoms by energy-adjusted dietary fibre intake, gender, and pandemic period (NHANES 2017–2023).

Variable	Category/Interaction	aPR (95% CI)†	p-value
Dietary fibre (per 5 g/day) (β_1)	–	0.80 (0.72, 0.89)	<0.001
Pandemic period (β_2)	Pandemic vs pre-pandemic	1.57 (1.23, 2.00)	0.001
Gender (β_3)	Female vs male	1.46 (1.07, 1.99)	0.017
Fibre $\times$ pandemic (β_4)	–	1.33 (1.13, 1.55)	0.001
Fibre $\times$ gender (β_5)	–	1.13 (0.99, 1.29)	0.072
Pandemic $\times$ gender (β_6)	–	0.77 (0.53, 1.13)	0.181
Fibre $\times$ pandemic $\times$ gender (β_7)	–	0.73 (0.58, 0.91)	0.007

†Model adjusted for race/ethnicity, age, education, marital status, income, BMI, sleep duration, medication use, physical activity, alcohol use, smoking, and hs-CRP; aPR: adjusted Prevalence Ratio; CI: Confidence Interval; NHANES: National Health and Nutrition Examination Survey.

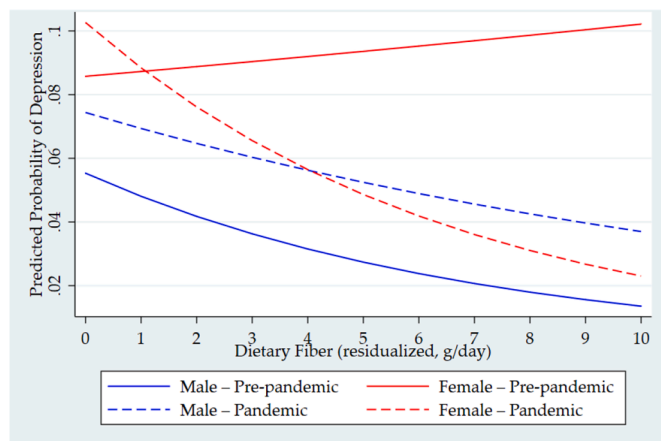


Fig. 3. Predicted marginal probabilities of depressive symptoms (PHQ-9 ≥ 10) by energy-adjusted dietary fibre intake stratified by gender and pandemic period, (NHANES 2017–2023).

Table 5

Association between energy-adjusted dietary fibre intake and depressive symptoms by gender and pandemic periods.

Gender	Period	Adjusted PR (95% CI)†	p-value
Male	Pre-pandemic	0.79 (0.71, 0.88)	<0.001
	Pandemic	1.04 (0.92, 1.17)	0.491
Female	Pre-pandemic	0.91 (0.84, 0.99)	0.038
	Pandemic	0.90 (0.81, 0.99)	0.037

†Model adjusted for age, race/ethnicity, education, marital status, income, BMI, sleep duration, prescribed medication use, vigorous physical activity, alcohol use, smoking status, and hs-CRP levels; PR: Prevalence Ratio; CI: Confidence Interval.

3.4. Subgroup analysis by gender

Table 5 presents the gender-stratified analyses. In the pre-pandemic period, higher energy-adjusted dietary fibre intake was associated with a lower prevalence of depressive symptoms in both men (aPR = 0.79; 95% CI: 0.71–0.88) and women (aPR = 0.91; 95% CI: 0.84–0.99). During the pandemic period, this association

was attenuated and centred around the null among men (aPR = 1.04; 95% CI: 0.92–1.17), whereas a modest inverse association persisted among women (aPR = 0.90; 95% CI: 0.81–0.99).

3.5. Sensitivity analyses

Sensitivity analyses supported the robustness of the main findings. First, complete-case estimates were broadly consistent with multiple-imputed models, although precision was reduced due to smaller sample sizes (see [Supplementary Table S2](#)). Second, re-categorising energy-unadjusted (absolute) fibre intake into quartiles produced similar patterns in pre-pandemic data, with higher energy-unadjusted fibre intake associated with lower prevalence of depressive symptoms. No clear dose–response pattern was observed during the pandemic period (see [Supplementary Table S3](#)). Third, to assess whether our main conclusions were influenced by the high proportion of missingness in smoking status (58.3%), we conducted a sensitivity analysis excluding current smoking status as a covariate ([Supplementary Table S4](#)). The key finding regarding the association between dietary fibre and depression remained virtually unchanged. The aPR for fibre in the pre-pandemic period was 0.86 (95% CI: 0.81, 0.92) in the main analysis and 0.86 (95% CI: 0.80, 0.92) in the smoking-excluded sensitivity analysis. During the pandemic, the aPR was 0.98 (95% CI: 0.91, 1.05) in the main analysis and 0.97 (95% CI: 0.90, 1.05) in the sensitivity analysis. The consistency of the estimates across models, despite the exclusion of a variable with 58.3% missingness, strongly suggests that our main findings are not an artifact of the imputation model for smoking status.

4. Discussion

Using nationally representative NHANES data, we examined whether the association between energy-adjusted dietary fibre and depressive symptoms differed among U.S. adults before and during the COVID-19 pandemic. Mean dietary fibre intake decreased from 16.52 g/day pre-pandemic to 15.95 g/day during the pandemic, remaining well below the recommended adult intake levels (25–38 g/day) [46]. Higher energy-adjusted dietary fibre intake was associated with a lower prevalence of depressive symptoms before the pandemic; however, this association was attenuated during the pandemic. A three-way interaction indicated that the inverse energy-adjusted dietary fibre–depression association persisted only among women. Consistent with this, gender-stratified analyses showed inverse associations in both men and women before the pandemic, but only among women during the pandemic, suggesting that the energy-adjusted dietary fibre–depression relationship varies by gender and contextual stressors.

Depressive symptom prevalence increased modestly from 8% pre-pandemic to 10.1% during the pandemic. Our estimates are consistent with those reported by Fan et al., who observed an increase in the prevalence of depression from 8.45% pre-pandemic to 12.2% during and after the pandemic [47], although our estimates are slightly lower. This difference likely reflects our more restrictive analytic sample, which required valid dietary recall data and complete covariate information, as well as the exclusion of implausible energy and fibre intakes. In addition, the use of multiple imputations may have reduced bias due to missing data but could also have favoured participants with healthier profiles, potentially leading to a modest underestimation of prevalence.

The attenuation of the association between energy-adjusted fibre intake and depressive symptoms during the pandemic likely reflects the impact of widespread societal stressors, including economic instability, social isolation, and pandemic-

related uncertainty [22,48–52]. Chronic stress can disrupt the HPA axis and alter gut microbiota composition, leading to reduced production of SCFA and impaired serotonergic signalling involved in mood regulation [17,19,21,53]. Concurrently, pandemic-related dietary shifts toward ultra-processed, energy-dense foods may have reduced overall diet quality, thereby diluting the protective association of fibre with mental health [54–56]. Elevated stress levels may also influence appetite regulation and food preferences, increasing the intake of high-fat and high-sugar foods that promote systemic inflammation and cardiometabolic dysregulation. Together, these behavioural and physiological disruptions may attenuate the modest protective effects of dietary fibre on depressive symptoms during the pandemic.

4.1. Gender differences in the fibre–depression association

A key finding of our study is the gender-specific pattern in the association between energy-adjusted dietary fibre intake and depressive symptoms across the two time periods. Among women, the inverse association between energy-adjusted dietary fibre and depression remained consistent across periods (~aPR 0.90). In contrast, among men, the association was evident in the pre-pandemic period and less pronounced during the pandemic. These findings are consistent with the established gender disparities in depression, whereby women demonstrate higher lifetime prevalence and greater inflammatory sensitivity [15,57,58].

Several biological mechanisms may explain this gender difference. Sex hormones, particularly oestrogen, influence immune function, stress reactivity, and gut–brain signalling pathways, potentially modifying the psychological response to dietary exposures. Oestrogen has been shown to modulate the HPA axis and enhance serotonergic neurotransmission, which may synergistically interact with fibre-derived SCFAs to promote mood stability [53,58]. Fibre exerts antidepressant-related effects through modulation of gut microbiota [18], enhancement of short-chain fatty acid production [11,13], and attenuation of systemic inflammation [38,59]—pathways that may operate differently in women due to sex-specific immune and endocrine profiles. Moreover, the gut microbiota composition differs between men and women [60], with potential implications for SCFA production efficiency and subsequent mental health effects.

Our findings are supported by prior literature demonstrating gender-specific associations between dietary fibre and depressive symptoms. Saghaian et al. reported that women in the highest quartile of fibre intake had 37% lower odds of depression, whereas no significant association was observed among men [61]. Similarly, Kim et al. found that increased fibre intake was inversely associated with depression among premenopausal women but not postmenopausal women [62], suggesting the potential role of hormonal modulation. Jian et al. demonstrated a non-linear inverse association between fibre intake and post-stroke depression among U.S. women [63]. In older adults, Chrzastek et al. observed that higher fibre intake was associated with fewer depressive symptoms in women, whereas in men, depressive symptoms were more strongly associated with higher sucrose intake [64]. Together, these findings reinforce the possibility of sex-dependent biological or behavioural pathways linking fibre intake to mental health outcomes.

The attenuation observed among men during the pandemic may reflect differential coping behaviours and contextual stress responses. Evidence suggests that men may be more likely to adopt maladaptive coping strategies under stress, including increased alcohol consumption and consumption of energy-dense ultra-processed foods [65–68]—behaviours that may counteract or overwhelm potential dietary fibre benefits. Men may also be more

susceptible to stress-induced alterations in the gut microbiota composition, though this requires further investigation. Moreover, acute and chronic stress may also override modest protective dietary effects through neuroendocrine and inflammatory activation. Thus, the unprecedented contextual stressors of the COVID-19 pandemic may have disproportionately attenuated the fibre-depression association among men, whereas women may have maintained some protective benefit through different physiological or behavioural pathways.

4.2. Strengths and limitations

This study has several strengths. We used nationally representative NHANES data spanning both pre-pandemic and pandemic periods, enabling direct comparison across two distinct contexts. The analysis incorporated comprehensive adjustment for potential confounders, sensitivity analyses to assess robustness of findings, and multiple imputations to minimise bias due to missing data. Moreover, we applied interaction and stratified analyses to examine the potential effect modification by gender and pandemic period. To further evaluate the robustness of our findings, we performed three sensitivity analyses using complete cases, quartile-based dietary fibre intake, and models excluding smoking status (Supplementary Tables S1–S3). The estimates were consistent across models, underscoring the robustness of the observed associations.

Several limitations should be considered. First, the cross-sectional design precludes the determination of causality or directionality; reverse causality is possible because individuals already experiencing depressive symptoms may consume a poorer-quality diet, including lower fibre intake. This reverse causality would bias associations away from the null if depression directly reduces fibre intake, or toward the null if the relationship is bidirectional. Second, dietary fibre intake was assessed using a single 24-h recall, which is subject to recall error and may not represent usual intake due to day-to-day variation. However, NHANES employs the validated Automated Multiple-Pass Method to enhance recall accuracy. Third, depressive symptoms were assessed through the self-reported PHQ-9 rather than a clinical diagnostic interview, which may introduce misclassification bias, although the PHQ-9 has demonstrated good validity and reliability in population-based studies [34]. Fourth, the pandemic period had lower response rates than previous cycles, which may introduce non-response bias. While we applied survey weights designed to mitigate this, potential for residual bias due to unmeasured differences between respondents and non-respondents remains. Finally, the COVID-19 infection status and specific types of fibre [69] (such as soluble and insoluble) were not available in the dataset and therefore could not be included in the analysis.

5. Conclusions

This population-based NHANES study demonstrates that higher energy-adjusted dietary fibre intake was associated with a modestly lower prevalence of depressive symptoms among U.S. adults before the COVID-19 pandemic, with protective associations observed in both men and women. During the pandemic, this association was attenuated overall and persisted only among women, suggesting that the dietary fibre–depression relationship appears to vary by gender and contextual stressors. These findings suggest gender differences in the persistence of the fibre–depression association under high-stress conditions, though the underlying mechanisms remain to be elucidated. Promoting fibre-rich diets remains a plausible and low-risk public health strategy for supporting mental well-being, particularly given that

the mean fibre intake in this population remained substantially below the recommended levels. Future longitudinal research incorporating biomarkers of inflammation and gut health, COVID-19 infection history, and diverse fibre sources is needed to elucidate mechanisms and inform targeted, potentially gender-specific, interventions for depression prevention.

Data availability

The data supporting this study are publicly available from the National Centre for Health Statistics (NCHS) via the National Health and Nutrition Examination Survey (NHANES) website (<https://www.cdc.gov/nchs/nhanes/>). All Stata code used for data management, statistical analysis, and figure generation is openly accessible at the GitHub repository: https://github.com/NebyuDanielAmaha/NHANES_Fiber.

Author contributions

Conceptualization, N.A.; methodology, N.A. E.Y.; software, N.A.; E.Y.; formal analysis, N.A.; writing—original draft preparation, N.A., K.B.; writing—review and editing, N.A., K.B., A.K. E.Y, Z.L; supervision, E.Y., Z.L. All authors have read and agreed to the published version of the manuscript.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clnesp.2026.103272>.

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